DIGITAL AND PRECISION AGRICULTURE

Overview

The Departments of Agronomy and Agricultural and Biosystems Engineering jointly offer a major leading to a Bachelor of Science (B.S.) degree in digital and precision agriculture (DPA). Graduates will possess knowledge of geospatial and sensor technologies, agricultural systems, data analysis, and modeling, enabling them to enhance the efficiency and sustainability of production systems. They will acquire the technical skills and fundamental knowledge necessary to support data-driven decisions that improve the economic, environmental, and social sustainability of agriculture. The program will take a systems thinking approach to empower graduates to maximize the value of data generated by new technologies and evaluate interactions between the environment, crops, and machines. By integrating modern precision agriculture practices on research/teaching farms, program participants can be directly immersed in their curriculum, get hands-on experience with production data, and create meaningful lives and livelihoods.

Graduates will be prepared for careers in precision farming, agricultural business and industry, crop production and soil management, yield forecasting, data analysis, agricultural machinery, agricultural service organizations, environmental and natural resource management, and research. The DPA degree also prepares students for graduate school.

Student Learning Outcomes

Upon graduation, students should be able to:

- demonstrate a solid understanding of fundamental agricultural principles and practices, including plant and soil science, crop production, pest management, and irrigation techniques necessary for utilizing digital and precision agriculture information.
- effectively utilize advanced technologies and tools in digital and precision agriculture, including geographic information systems (GIS), remote sensing, global positioning systems (GPS), drones, and data analytics.
- successfully collect, manage, analyze, interpret, and communicate agricultural data effectively, including data collection methods, data quality assessment, statistical analysis, and interpretation of results for informed decision-making.
- develop a dynamic, context-specific decision framework that utilizes problem-solving and critical thinking skills to integrate agronomic knowledge with digital and precision agriculture technologies, addressing challenges in real farming scenarios.

Curriculum in Digital and Precision Agriculture

Total Degree Requirements: 120 cr.

Only 65 cr. from a two-year institution may apply which may include up to 16 technical cr.; 9 P-NP cr. of free electives; 2.00 minimum GPA.

International Perspective: 3 cr.

3 cr. from approved International Perspective list: http://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current (http://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current/) (AGRON 1800 recommended).

U.S. Cultures and Communities: 3 cr.

3 cr. from approved U.S. Cultures and Communities list: http://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses (https://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses/)

Communication/Library 13 cr.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
LIB 1600	Introduction to College Level Research	1
AGEDS 3110	Presentation and Sales Strategies for Agricultural Audiences	3
One of the following:		
ENGL 3090	Proposal and Report Writing	3
ENGL 3120	Communicating Science and Public Engagement	3
ENGL 3140	Technical Communication	3

Humanities: 3 cr.

3 cr. from approved humanities list: http://www.cals.iastate.edu/student-services/humanities (http://www.cals.iastate.edu/student-services/humanities/)

Social Sciences: 3 cr.

3 cr. from approved social sciences list: http://www.cals.iastate.edu/student-services/social-sciences (http://www.cals.iastate.edu/student-services/social-sciences/) (SOC 4150 Recommended)

Ethics: 3 cr.

One of the following:		
AGRON 3420	World Food Issues: Past and Present	3
AGRON 4500	Issues in Sustainable Agriculture	3
SOC 3250	Agriculture in Transition	3
SOC 3620	Applied Ethics in Agricultural and Rural Development	3

Life Sciences: 3 cr.

BIOL 2510	Biological Processes in the Environment	3

Physical Sciences: 8 cr

AGRON 2060	Introduction to Weather and Climate	3
One of the following:		
CHEM 1630 & 1630L	College Chemistry and Laboratory in College Chemistry	5
CHEM 1670 & 1670L	General Chemistry for Engineering Students and Laboratory in General Chemistry for Engineering	5
CHEM 1770 & 1770L	General Chemistry I and Laboratory in General Chemistry I	5

Mathematical Sciences: 7 cr.

One of the following:		
MATH 1430	Preparation for Calculus	4
MATH 1600	Survey of Calculus	4
MATH 1650	Calculus I	4
One of the following:		
STAT 1010	Principles of Statistics	4
STAT 1040	Introduction to Statistics	3

Digital Precision Agriculture Core: 36 cr.

DPA 1100	Professional Development in Digital and Precision Agriculture: Orientation	1
DPA 2020	Introduction to Digital and Precision Agriculture	3
TSM 1150	Solving Technology Problems	3
COMS 1270	Introduction to Computer Programming	3
AGRON 1810	Introduction to Crop Science	3
AGRON 1820	Introduction to Soil Science	3
AGRON 2700	Geospatial Technologies	3
STAT 3201	Intermediate Statistical Concepts and Methods	4
or MATH 1650	Calculus I	
ENSCI 3180	Introduction to Ecosystems	3
or BIOL 3120	Ecology	
ECON 3340	Entrepreneurship in Agriculture	3
AGRON 4020	Digital and Precision Agricultural Systems	4
TSM 4330	Precision Agriculture	3
DPA 3100	Professional Development in Digital and Precision Agriculture: Work Experience	R

Focus Area: 21 cr.

- Complete required 12 credits from one focus area.
- Complete 9 additional credits from chosen focus area, or any focus area.

Electives: 17 cr.
Total: 120 cr.

Focus Areas:

Precision Agronomy Focus Area

Required Courses:		
AGRON 2800	Crop Development, Production and Management	3
AGRON 3540 & 3540L	Soils and Plant Growth and Soils and Plant Growth Laboratory	4
NREM 4890	Survey of Remote Sensing Technologies	3
or CRP 4540	Fundamentals of Remote Sensing and Spatial Analysis	
AGRON 3920	Systems Analysis in Crop and Soil Management	3
8 credits from the following or other options:		
TSM 2100	Fundamentals of Technology	3
ENT 3760	Fundamentals of Entomology and Pest Management	3
PLP 4080	Principles of Plant Path	3
AGRON 3170	Principles of Weed Science	3
ECON 2360	Agricultural Selling	3

Applied Data Management Focus Area

Required Courses:		
DS 2010	Introduction to Data Science	3
DS 2020	Data Acquisition and Exploratory Data Analysis	3
AGRON 4250	Crop and Soil Modeling	3
COMS 2280	Introduction to Data Structures	3
9 credits from the following or other options:		
COMS 3630	Introduction to Database Management Systems	3
DS 3010	Applied Data Modeling and Predictive Analysis	3
STAT 3132	Visual Communication of Quantitative Information	3

Geospatial Applications Focus Area

Required Courses:		
NREM 4460	Integrating GPS and GIS for Natural Resource Management	3
STAT 4206	Spatial Data Analysis	3
CRP 4540	Fundamentals of Remote Sensing and Spatial Analysis	3
CRP 4560	GIS Programming and Automation	3
9 credits from the following or other options:		
NREM 4890	Survey of Remote Sensing Technologies	3
GEOL 4880	Raster GIS for Geoscientists	3
DS 2010	Introduction to Data Science	3

Precision Conservation Focus Area

Required Courses:		
AGRON 4250	Crop and Soil Modeling	3

STAT 4206	Spatial Data Analysis	3
TSM 3240	Soil and Water Conservation Management	3
or AGRON 3600	Environmental Soil Science	
or AGRON 3920	Systems Analysis in Crop and Soil Management	
NREM 4600	Controversies in Natural Resource Management	3
9 credits from the following or other options:		
CRP 4540	Fundamentals of Remote Sensing and Spatial Analysis	3
BIOL 3190	Analysis of Environmental Systems	3
BIOL 3700	GIS for Ecology and Environmental Science	1-6
BIOL 4830	Environmental Biogeochemistry	3
BIOL 4850	Community Ecology	3

Ag Technology Focus Area

Required Courses:		
TSM 2100	Fundamentals of Technology	3
TSM 3300	Agricultural Machinery and Power Management	3
TSM 3630	Electrical Power and Electrical Machinery Systems	3
TSM 3370	Fluid Power Systems Technology	3
9 credits from the following or other options:		
TSM 3350	Tractor Power	4
TSM 3650	Electrical Controls and Sensor Systems	3
DS 2010	Introduction to Data Science	3
NREM 4890	Survey of Remote Sensing Technologies	3
or CRP 4550	Smart and Sustainable Cities	
TSM 2160	Advanced Technical Graphics, Interpretation, and CAD	2

Four Year Plan

Freshman

Fall	Credits	Spring	Credits
DPA 1100	1	AGRON 1810	3
AGRON 1800	3	TSM 1150	3
ENGL 1500	3	AGRON 1820	3
LIB 1600	1	COMS 1270	3
MATH 1430, 1600, or 1650	4	STAT 1010 or 1040	3-4
CHEM 1630	4		
CHEM 1630L	1		
	17		**15-16**

Sophomore

Fall	Credits	Spring	Credits	Summer	Credits
AGRON 2060	3	STAT 3201 or MATH 1650	4	DPA 3100	Required
DPA 2020	3	Humanities	3		
DPA Specialization Course	3	AGRON 2700	3		
ECON 1010	3	BIOL 2510	3		
ENGL 2500	3	DPA Specialization Course	3		
	15		**16**		**0**

Junior

Fall	Credits	Spring	Credits
Elective	3	AGRON 3420, 4500, or SOC 3620	3
Elective	3	ENSCI 3180 or BIOL 3120	3-4
AGEDS 3270, ENGL 3090, ENGL 3020, or ENGL 3140	3	SOC 3250	3
ECON 3340	3	DPA Specialization Course	3
AGEDS 3110	3	Elective	3
	15		**15-16**

Senior

Fall	Credits	Spring	Credits
Elective	3	DPA Specialization Course	3
Elective	3	DPA Specialization Course	3
DPA Specialization Course	3	US Cultures and Communities	3

DPA Specialization Course	3	DPA 4020	4
TSM 4330	3	Elective	3
	15		**16**